Fabio Vulpi

Ph.D. in Mechanical Engineering and Management

+39 3803485030 | 70124, Bari (BA), Italy | 21/12/1995
linkedin.com/in/fabio-vulpi | github.com/vulpi-f | vulpi.fabio@gmail.com

ABOUT ME

Daring and versatile, I firmly believe that attaining successful outcomes relies on both the ability to ask pertinent questions and the commitment to finding the answers.

SOFT-SKILLS

- Pragmatic
- Team working
- Result Oriented

DIGITAL SKILLS

Physic Simulation

MATLAB | Ansys FLUENT
Simulink | Wolfram Mathematica

μC programming

Debian | Python | C | Arduino
Linux | ROS | C++ | C#

3-D Modeling

AutoCAD 2D | Autodesk Invetor
SolidWorks | Cura (Ultimaker)
MeshLab | Eagle - PCB design

LANGUAGES

Italiano — C 2
English — C 1
Deutsch — A 2

PUBLICATIONS

- Kalman Supervised Network for improved Model Predictions.
- Recurrent and convolutional NN for deep terrain classification by autonomous robots.
- An RGB-D multi-view perspective for autonomous agricultural robots.
- On the role of feature and signal selection for terrain learning in planetary exploration.
- Usability Study of WISE for Movement Assessment and Exercise.
- Wearable Inertial Sensors for Range of Motion Assessment (WISE).

WORK EXPERIENCE

Electro-Avionics Engineer — 02/2026 - current

Unmanned Aero Systems s.r.l. - Brindisi (IT)

Sized, selected, and integrated UAV avionics and control electronics, including datalinks, servo actuators, IMUs, sensors, and onboard control units. Configured and tuned ArduPilot autopilots, developed mission-specific flight-control logic, and customized QGroundControl-based ground stations for telemetry, mission management, and UAV control.

Experienced Consultant — 10/2024 - 02/2026

Amaris Consulting - Turin (IT)

Vehicle Functional Safety (FuSa) consultant for ISO 26262:2018. Supported Safety Concept and Development Interface Agreement (DIA) definition, led HARA/FMEA reviews, and validated safety requirements across the automotive safety lifecycle.

Software System Engineer — 01/2023 - 10/2024

Bosch CVIT - Bari (IT)

Served as a Functional Safety Software Engineer on Engine and Battery Control Unit V-model development, ensuring ISO 26262 compliance across diesel, electric, light- and heavy-duty platforms. Engineered AI/ML convenience features such as passive keyless entry and gesture-based trunk release using signal processing and device-tracking models.

Software Automation Engineer — 11/2022 - 01/2023

Links Management & Technology - Bari (IT)

Built AI/ML software solutions to optimize automation workflows.

EDUCATION

Ph.D. in Mechanical Engineering and Management — 11/2019 - 04/2023

Polytechnic of Bari - Bari (IT)

Collaborated with CNR-STIIMA on EU-funded projects ADE, ATLAS, and ANTONIO, leading HW/SW development and AI/ML algorithm design for robotic systems.

- ADE: Built deep-learning terrain-classification pipelines for exploration rovers.
- ATLAS: Engineered a ROS-based, multi-sensor platform for 3D field reconstruction.
- ANTONIO: Developed signal- and image-processing workflows for crop-monitoring AGRs.

Master of Science in Mechanical Engineering — 10/2017 - 10/2019

NYU - New York (USA) — 110L/110 - GPA 3.952/4

Completed a double-degree M.Sc. in Mechatronics and Robotics with the Polytechnic of Bari. Authored a Medical Engineering thesis on wearable IMU systems for quantitative upper-limb range-of-motion assessment.

Bachelor Degree in Mechanical Engineering — 07/2014 - 07/2017

Polytechnic of Bari - Bari (IT) — 109/110

Performed physics-based modeling and numerical computation to investigate the nonlinear dynamics of oscillators with a bistable potential.

AWARDS

Ingeniarius Ltd - Porto (PT) — RobotCraft First Place Award - 08/2022

Designed and assembled an autonomous mobile robot to explore a maze, generate LiDAR-based SLAM maps in ROS, and re-navigate through optimized paths. Developed real-time C++/Python mapping and planning algorithms and integrated Arduino-driven motor control.